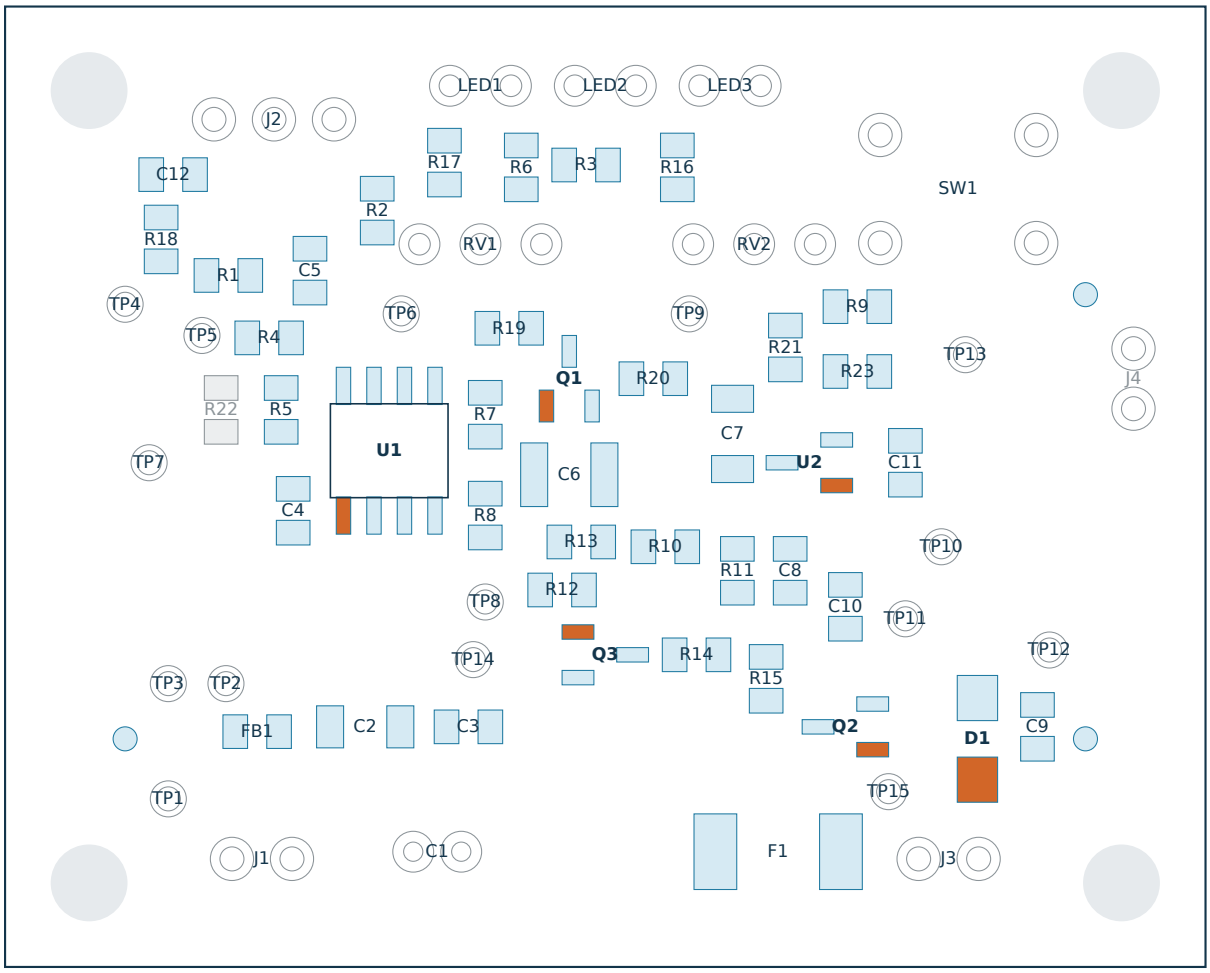
BUBBLELINK / CTRL v1.2

Assembly and polarity reference

41 top-side SMT placements. THT, test points, mounting holes and DNP parts are not assembly items.

BOARD TOP / 50 x 40 mm



Orange land = physical pad 1 for U1, U2, Q1-Q3 and D1.

D1 pad 1 is cathode: align the package band with the orange land.

Datum: lower-left corner. +X right, +Y up. Do not mirror.

Critical orientation checks

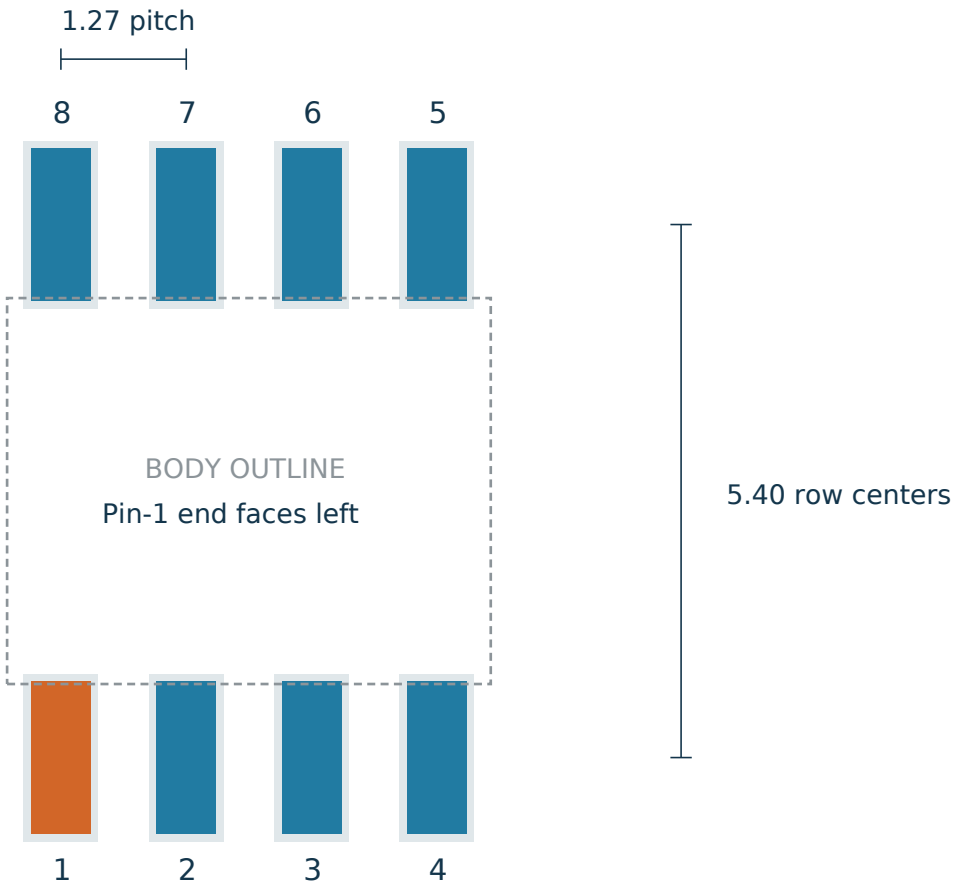
Ref	Pad 1 function	X (mm)	Y (mm)
U1	OUT1	14.0950	18.8000
U2	GND	34.6375	20.0500
Q1	Gate	22.5500	23.3625
Q2	Gate	36.1375	9.0500
Q3	Gate	23.8625	13.9500
D1	Cathode / band	40.5000	7.8000

J4 and R22: DNP. Fit C9 and R18. No through-hole or bottom assembly.

Use the supplied BOM, CPL and assembly notes. Numeric rotations alone are insufficient.

U1 / TI D0008A land pattern

Keep Texas Instruments TLV1822QDRQ1. This revision corrects the PCB footprint; no electrical substitution.



Copper and paste: 8 x 1.55 across the rows x 0.60 along each row.

Mask opening: 0.75 x 1.70; 0.075 expansion per side. Non-mask-defined lands.

Copper inner gap 3.85; outer span 6.95. Lead pitch is 1.27, not 5.40.

Numbered pad coordinates / finished-board lower-left datum

Pad	Function	X (mm)	Y (mm)
1	OUT1	14.0950	18.8000
2	IN1-	15.3650	18.8000
3	IN1+	16.6350	18.8000
4	V- / GND	17.9050	18.8000
5	IN2+	17.9050	24.2000
6	IN2-	16.6350	24.2000
7	OUT2	15.3650	24.2000
8	V+	14.0950	24.2000

Assembly stencil reference: TI example uses 0.125 mm thickness. PCBWay should confirm its process choice and paste release for the complete SMT population.

Source: TI D0008A, drawing 4214825/C, board-layout and stencil examples.